Projected Gradient Descent and Frank-Wolfe

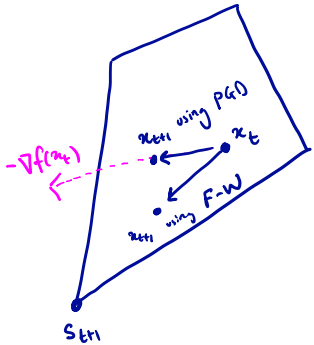
$$\min: f(x)$$
$$\text{st}: x \in \mathcal{X}$$

PGD:
$$\left.\begin{aligned} y_{t+1} &= x_t - \eta \nabla f(x_t) \\ x_{t+1} &= \text{Proj}_{\mathcal{X}}(y_{t+1}) \end{aligned}\right] \quad x_{t+1} = \text{Proj}_{\mathcal{X}}(x_t - \eta \nabla f(x_t))$$

Frank-Wolfe:
$$s_{t+1} \in \arg\min: \nabla f(x_t)^T s$$
$$\text{st}: s \in \mathcal{X}$$

$$x_{t+1} = (1-\gamma_t) x_t + \gamma_t s_{t+1}$$

$$\gamma_t = \frac{2}{t+1}$$



PGD vs Frank-Wolfe

Why would we not simply follow $-\nabla f$?

(1) Projection may be expensive —

In some cases projection step is more expensive than solving the min in FW $\left(\begin{array}{l} \min: \nabla f(x_t)^T s \\ \text{st: } s \in \mathcal{X} \end{array} \right)$.

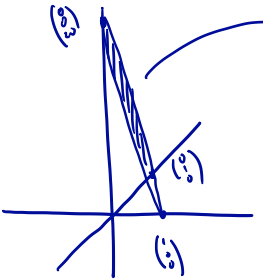
(2) $\longrightarrow$ see example.

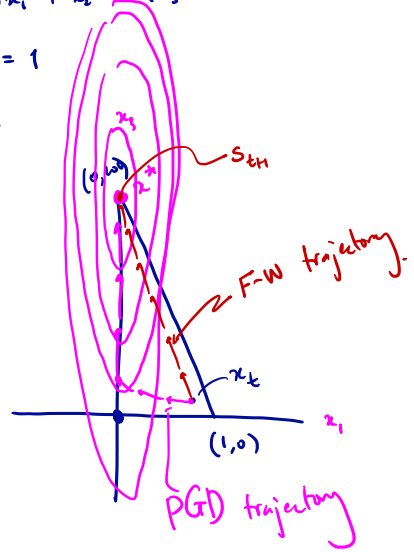
Frank-Wolfe Example — 1/2

$$\min: \; f(x_1, x_2, x_3) = 100x_1^2 + x_2^2 + (x_3 - 20)^2$$

$$\text{st}: \; x_1 + x_2 + x_3/20 = 1$$

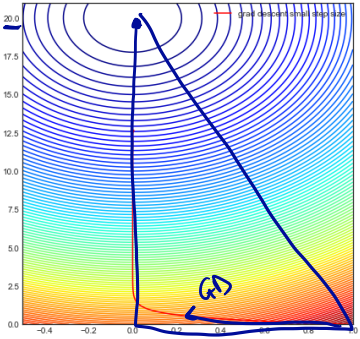
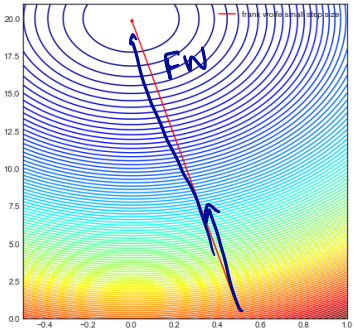
$$x_1, x_2, x_3 \geq 0.$$





Frank-Wolfe Example — 2/2

Gradient Descent



Frank-Wolfe